Huihan Li

huihanlhh.github.io

USC PhD Candidate in Computer Science, expected graduation May 2027 Los Angeles, CA

Summary: Researcher making LLM reasoning reliable and interpretable. I build diagnostic tools that trace model failures to inspectable causes in reasoning steps or training data and use them to target interventions. In product work I apply this evidence-first rigor, defining measurable product criteria for stochastic agents at enterprise scale under sparse data.

Research and Industry Experience

Adobe, Adobe Experience Platform San Jose, CA
Applied Scientist Intern, Agent & GenAI, with Yunyao Li *May 2026 – Aug 2026*

- Led the redesign of the orchestration layer of Adobe's enterprise agent platform (∼300 capabilities), from requirements with 7 product teams to production deployment. Patent application submitted.
- Applied progressive disclosure and self-correcting capability descriptions, raising routing reliability from 56% to 75% while staying robust to catalog growth and modification.
- Established gold labels and success criteria for non-deterministic agent trajectories and built the platform's first routing benchmark (∼400 queries). Adopted by 2+ other agent projects as a shared evaluation set.
- Built the offline evaluation and stress-testing pipeline that estimates user-facing impact of orchestration changes before release, validated on production traffic under sparse enterprise data.

Meta, GenAI Bellevue, WA
Research Intern, with Liang Tan *May 2025 – Aug 2025*

- Studied dynamic temperature scheduling for RL post-training of reasoning models: extended VERL to vary rollout temperature over training for GRPO, post-trained Qwen3-1.7B, and designed controlled evaluations on AIME and MATH with pass@1, pass@16, and problem-level analysis.
- Characterized a precision-diversity tradeoff: sinusoidal schedules improve multi-sample success more than single-sample accuracy, and dynamic-temperature GRPO offers no gain over constant-temperature training.

Allen Institute for AI, Mosaic Seattle, WA
Research Intern, with Nouha Dziri and Yejin Choi *May 2023 – Mar 2024*

- Led the CULTURE-GEN project (COLM 2024), designing a culture-conditioned generation framework that measures how LLMs represent 110 countries and regions across 8 topics and exposes systematic disparities between marginalized and default cultures. Findings motivated follow-up work on pretraining-data attribution (ICLR 2025).

Additional experience: Apple AI/ML (*MLE Intern*, multilingual question answering and JAX/Flax, 2022); Google Assistant (*SWE Intern*, dialog systems and natural language understanding, 2019).

Selected First-Author Publications

- **ARBOR: Locating Reasoning Failures in Long Chains of Thought through Reasoning State Tracking** *arXiv, 2026.* A tree-structured engine that tracks thought lineage in long CoT and localizes repetition, underthinking, and unsupporte deduction failures to the relation between a step and the state preceding it. [paper] [code]
- **Diagnosing Memorization in Chain-of-Thought Reasoning, One Token at a Time** *EMNLP 2025.* Token-level attribution of chain-of-thought steps to memorized pretraining content. [paper] [code]
- **Attributing Culture-Conditioned Generations to Pretraining Corpora** *ICLR 2025.* Traced culture-conditioned generations to pretraining data at corpus scale. [paper] [code]
- **CULTURE-GEN: Revealing Global Cultural Perception in Language Models through Natural Language Prompting** *COLM 2024.* Evaluated global cultural perceptions in LLMs. [paper] [code]
- **Systematic Generation of Long-Tail Inferential Knowledge via Logical Rule Guided Search** *EMNLP 2024.* Generated long-tail inferential knowledge through logical-rule-guided search. [paper] [code]
- **Ditch the Gold Standard: Re-evaluating Conversational Question Answering** *ACL 2022, Outstanding Paper Award.* Evaluated conversational QA beyond a single gold answer. [paper] [code]

Education

University of Southern California, PhD in Computer Science; Advisor: Xiang Ren; Amazon Fellow 2022 – Pres.
Princeton University, MSE in Computer Science; Advisor: Danqi Chen; Siebel Scholar 2020 – 2022
Wellesley College, BA in Computer Science and Cognitive and Linguistic Sciences; *magna cum laude* 2016 – 2020

Technical Skills

Python, PyTorch, Hugging Face, vLLM, VERL, JAX/Flax, FSDP, Ray, CUDA, Slurm, Weights & Biases, Pandas, Elasticsearch.